

CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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ABSTRACT

We study whether vision-language models can execute declared classical search algorithms rather than directly generate complete plans. A Search Process Policy emits a Typed Search Operation whose explicit operands determine the next exploration decision. A Trusted Search Runtime validates and applies the emission, maintains bounded external Search Memory, checks Algorithm Invariants, and charges invalid operations without choosing omitted operands or silently repairing outputs. This contract separates stepwise validity from full-episode structural/process competence. In a frozen, outcome-blind 15-task BFS development panel, final process-SFT checkpoints from five training seeds achieved 1.0 invariant-valid success with zero invalid operations; the pretrained base achieved 0.0, while a random-valid reference also achieved 1.0. The learned policy therefore showed executable BFS behavior on the selected panel but no gain over the strongest control, producing a governed `VALID_STOP` with `scientific_completion=false`. The next structural stage uses a complete unpruned BFWS variant: exact traces and a released text-state corpus now exist, but training and adjudication remain incomplete. A*, visual-state, multimodal-state, DAgger, and final comparative evaluation remain prospective.

1 INTRODUCTION

Classical planning requires finding a sequence of actions that transforms an initial state into a goal state under specified constraints—a search problem whose formal structure allows instances to be generated systematically, solutions verified automatically, and difficulty controlled precisely, making it a natural probe of reasoning in foundation models (Valmeekam et al., 2023; Stein et al., 2025; Kokel et al., 2026). The formal model behind each instance fixes the task semantics, so errors reflect reasoning rather than ambiguity in the problem statement. The explicit state space also supports two levels of question: how actions apply locally, and how states connect globally. These properties let an evaluator vary difficulty along controlled dimensions and check each answer mechanically.

A recent study used this setting to measure latent planning competence in off-the-shelf language models (Huang et al., 2026). It separated two competencies that did not compensate for each other. Operational reasoning covers local applicability, progression, and validation: whether an action is legal in a state, what state it produces, and whether a goal condition holds. Structural enumeration covers goal reachability, landmarks, and other global state-space structure. Across model scales, and under chain-of-thought prompting, targeted prompting, and scratchpad or agent harnesses, operational reasoning improved, but these gains were not accompanied by comparable structural gains. The evidence was item-level and text-only, drawn from models used as released, without task-specific training.

Structural enumeration is related to global search-space reasoning, but it is not identical to full search execution. Identifying which states are reachable, or which serve as landmarks, does not require controlling a frontier across a budgeted episode. The prior study established this profile observationally. It did not test whether the structural competency can be acquired through training, or whether a model can sustain a declared procedure over a complete episode. Closing that gap calls for direct intervention: train the model to make the exploration decisions of a classical search algorithm, then measure whether structural/process competence changes.

Our intervention replaces direct plan generation with learned execution of a declared search algorithm. A Search Process Policy is trained to emit Typed Search Operations, discrete steps such as expanding a state, generating a successor, updating the frontier, or applying a goal test. A Trusted Search Runtime validates and applies each emission, checks Algorithm Invariants on every operation, and maintains bounded external Search Memory holding frontier, visited, novelty, and priority state. Ownership is replay-determinative. The emission must contain every operand that determines exploration, and the runtime may not select omitted operands, reorder candidates for the policy, or silently repair an invalid emission. Replaying the raw emissions against the same task and Search Memory must reproduce the explored trace. Operation validity, invariant compliance, episode success, and learned advantage over uninformed exploration are measured separately.

The study asks whether a policy can learn to execute declared algorithms with valid, invariant-compliant traces, whether learned execution improves structural/process competence beyond the valid but uninformed exploration of a random-valid reference, and whether the acquired capability transfers to planning benchmarks and broader search-intensive reasoning. The design spans BFS, BFWS, and A* under $h_{\max}$ and landmark-count heuristics. Training uses process supervision on search-trace segments, with DAgger as a prospective refinement. Modality conditions present the same task content as text-state, visual-state, or multimodal-state observations under equal Search Memory capacity. Reference conditions include the pretrained base model and exact classical execution. Beyond the BFS stage, these components remain prospective.

One comparison is complete. On a frozen 15-task BFS development panel, the process-supervised policy solved every task with no invalid operations, the pretrained base solved none, and the random-valid reference also solved every task. Executable BFS behavior is therefore observed for the trained policy on this panel, but with no learned advantage over uninformed valid exploration. After width-based attempts proved incomplete as training data, the structural stage moved to a complete unpruned BFWS variant. Exact BFWS traces and a released text corpus are now complete, while BFWS training and adjudication remain unfinished. A*, the modality conditions, DAgger, the final comparative evaluation, and all transfer questions remain future work.

This paper makes three contributions.

- **From diagnosis to intervention.** We formulate the training-based follow-up to a diagnosed structural bottleneck: rather than prompt around flat structural enumeration, we test whether the capability can be acquired as executable search under a declared algorithm.
- **A replay-determinative policy/runtime interface.** Typed Search Operations carry every exploration-determining operand, while the no-repair Trusted Search Runtime validates and applies them. Operation validity, invariant compliance, episode success, and learned control advantage remain separate measurements.
- **A staged empirical framework with a bounded first result.** The framework pairs per-algorithm trace corpora with frozen evaluation panels. Its completed BFS stage shows executable learned behavior without advantage over random-valid exploration, and its BFWS stage contributes completed traces and a text corpus with training pending.

The paper therefore pairs a bounded negative BFS comparison with completed BFWS infrastructure and a prospective broader evaluation, and the remaining sections specify the operation contract, the accumulated evidence, and the conditions required before wider claims can be made.

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A APPENDIX

The Appendix will retain the complete evidence contract, including operation schemas, Algorithm Invariant definitions, runtime pseudocode, data and split procedures, receipt tables, statistics, adapters, and narrowly scoped feasibility details of retained infrastructure.